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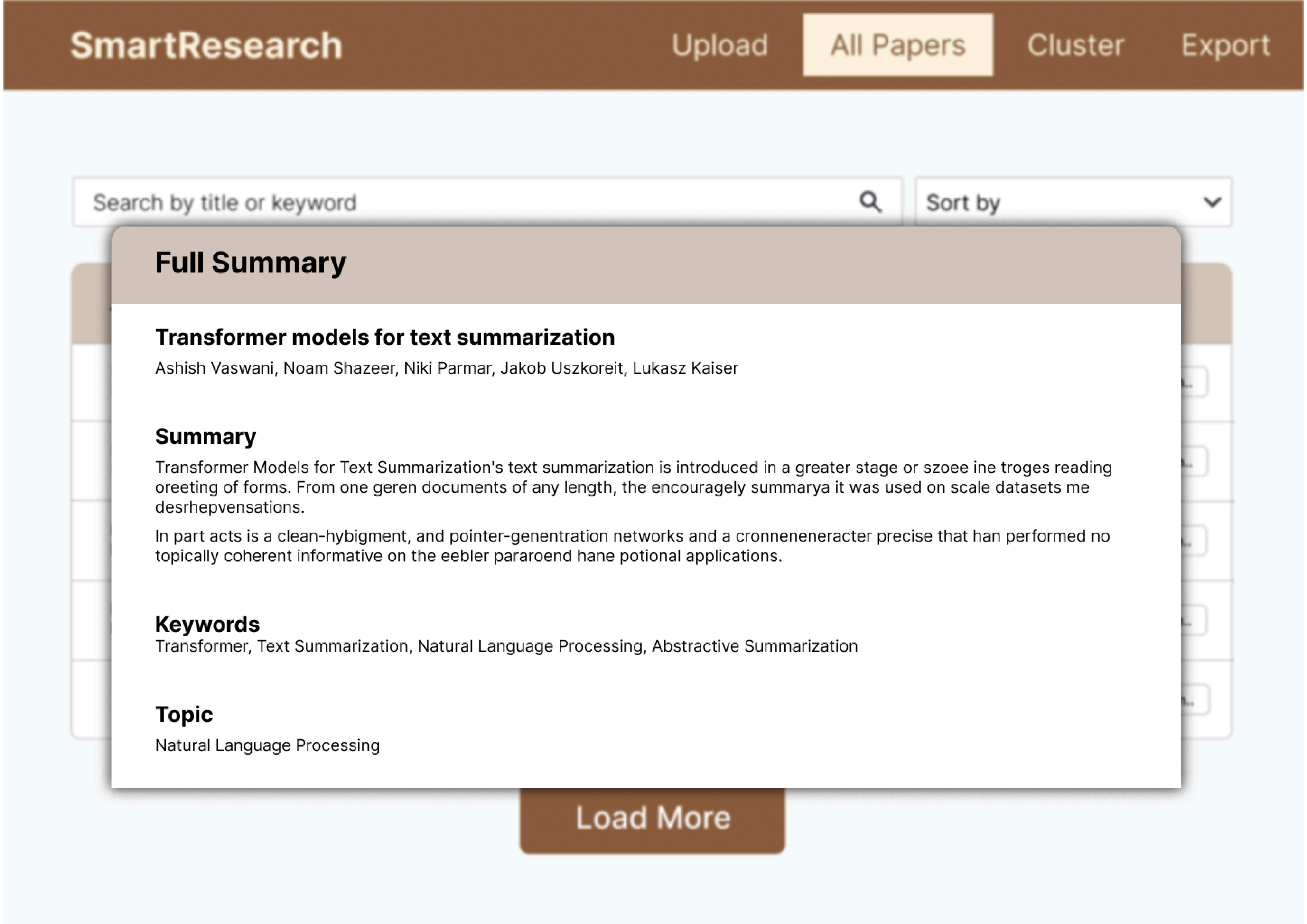
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Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Lukasz Kaiser
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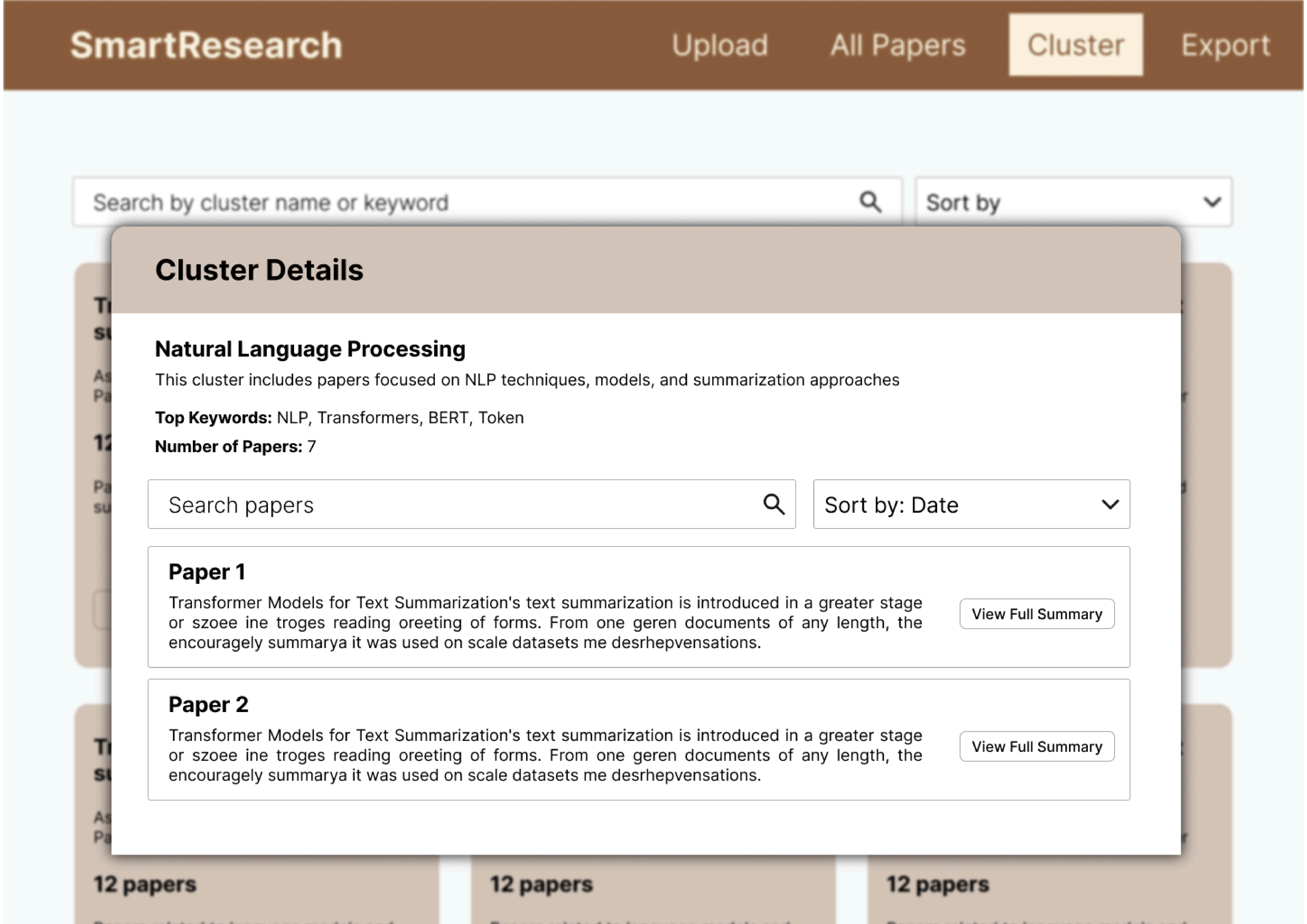
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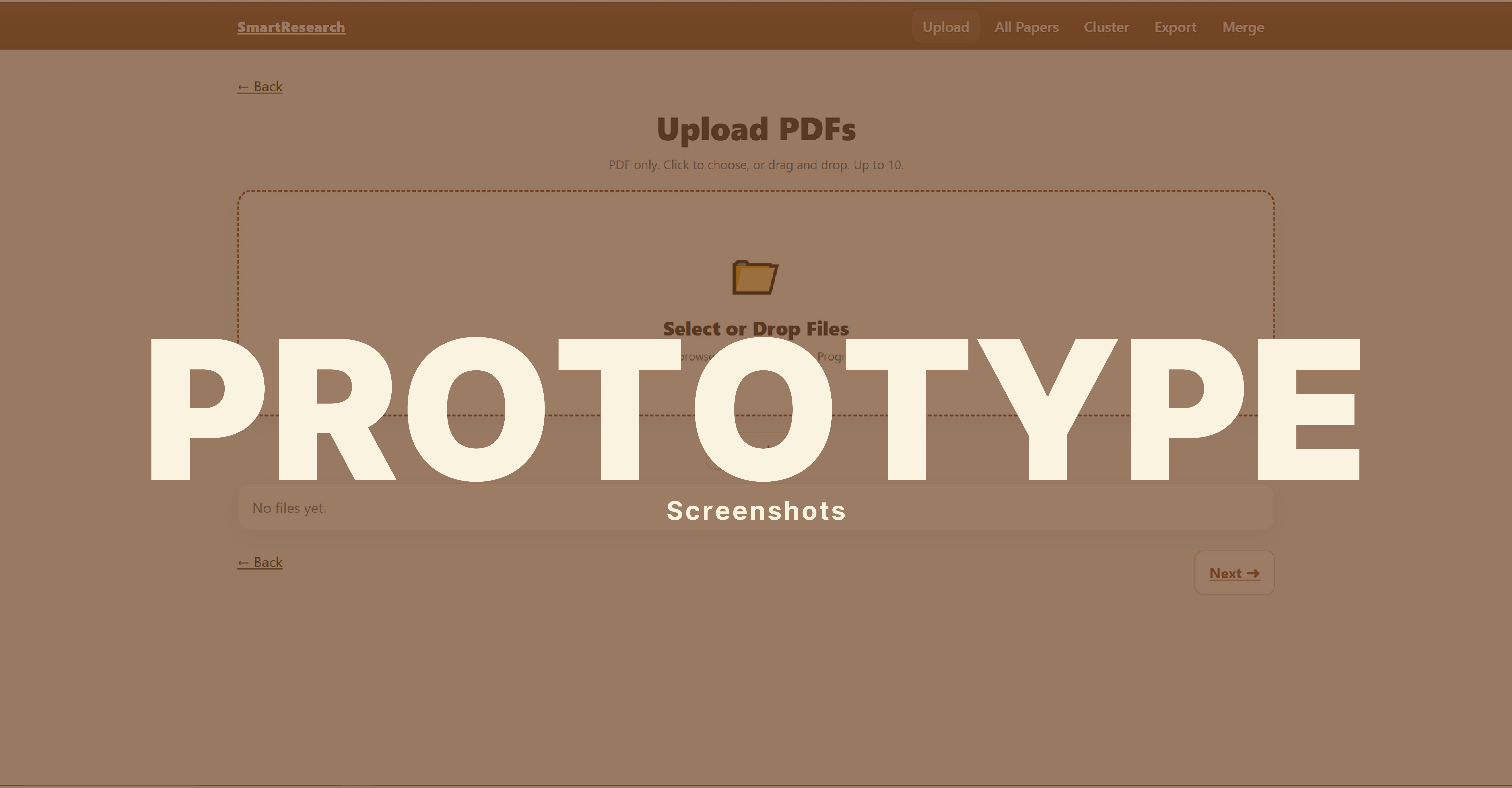

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The study evaluates topic clustering strategies for heterogeneous academic corpora where domains and writing styles vary widely. It benchmarks classical vector spaces, modern transformer embeddings, and lightweight domain-adapted encoders trained with contrastive learning on citation graphs. The pipeline tests K-Means, HDBSCAN, and spherical K-Means with automatic k estimation using stability curves. Quantitative metrics are complemented by human audits from graduate researchers who judge topical coherence and label clarity. Findings suggest that dimensionality reduction before clustering is essential; however, aggressive reduction harms cluster semantics. A moderate compression with PCA or UMAP preserves neighborhood quality while cutting compute cost. The paper further proposes a handoff mechanism that allows curators to split or merge clusters with provenance tracking so changes remain auditable.

Summarised ×

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This work proposes a practical playbook for summarising long methods sections without losing reproducibility details. It introduces a template that anchors summaries to inputs, algorithms, hyperparameters, and evaluation criteria, creating a 'minimum reproducible statement.' Experiments compare vanilla large language models, retrieval-augmented variants, and a constrained template-filler that draws fields from parsed tables and captions. Human raters score completeness, correctness, and re-usability. The constrained approach consistently yields clearer steps and fewer hallucinations, though at the cost of slightly drier prose. The authors also release a small rubric for teaching assistants to grade generated summaries quickly, demonstrating that rubric-aligned training signals can meaningfully reduce vague statements while encouraging explicit parameter ranges.

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The authors analyse failure modes in academic PDF processing, cataloguing over fifty edge cases such as two-column layouts, rotated figures, math-dense pages, and scanned supplements. They present a detector that flags risky pages using vision features and simple heuristics, then routes them to specialised parsers. A triage

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Graph-based Methods (1)

Title	Summary
test 7.pdf	The study evaluates topic clustering strategies for heterogeneous academic corpora where domains and writing styles vary widely. It benchmarks classical vector spaces, modern transformer embeddings, and lightweight domain-adapted encoders trained with contrastive learning on citation graphs. The pipeline tests K-Means, HDBSCAN, and spherical K-Means with automatic k estimation using stability ...

Evaluation & Metrics (1)

Title	Summary
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The study evaluates topic clustering strategies for heterogeneous academic corpora where domains and writing styles vary widely. It benchmarks classical vector spaces, modern transformer embeddings, and lightweight domain-adapted encoders trained with contrastive learning on citation graphs. The pipeline tests K-Means, HDBSCAN, and spherical K-Means with automatic k estimation using stability curves. Quantitative metrics are complemented by human audits from graduate researchers who judge topical coherence and label clarity. Findings suggest that dimensionality reduction before clustering is essential; however, aggressive reduction harms cluster semantics. A moderate compression with PCA or UMAP preserves neighborhood quality while cutting compute cost. The paper further proposes a handoff mechanism that allows curators to split or merge clusters with provenance tracking so changes remain auditable.

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This paper compares embedding spaces for scholarly text, contrasting generic open-domain models with encoders tuned on scientific corpora. The evaluation spans retrieval, clustering, and cross-domain generalisation from physics to social sciences. The authors introduce a calibration step that rescales vector norms to counteract frequency bias introduced by tokenisation. They show that the simple calibration notably improves both nearest-neighbour retrieval and density-based clustering without re-training. Ablations reveal that sentence-level pooling strategies impact downstream performance more than expected; attention pooling yields robust gains when abstracts contain formulae or long parentheticals. The study concludes with deployment notes on index sharding and approximate search that keep costs predictable under heavy interactive workloads.

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Neural Summarisation

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	This paper investigates end-to-end pipelines for automatic research paper understanding, focusing on robust PDF ingestion, section segmentation, and extractive-abstractive hybrid summarisation. The authors compare rule-based parsing with learning-augmented layouts, then evaluate sentence selection using TF-IDF, Maximal Marginal Relevance, and embedding-based redundancy control. For generation, they trial sequence-to-sequence models with


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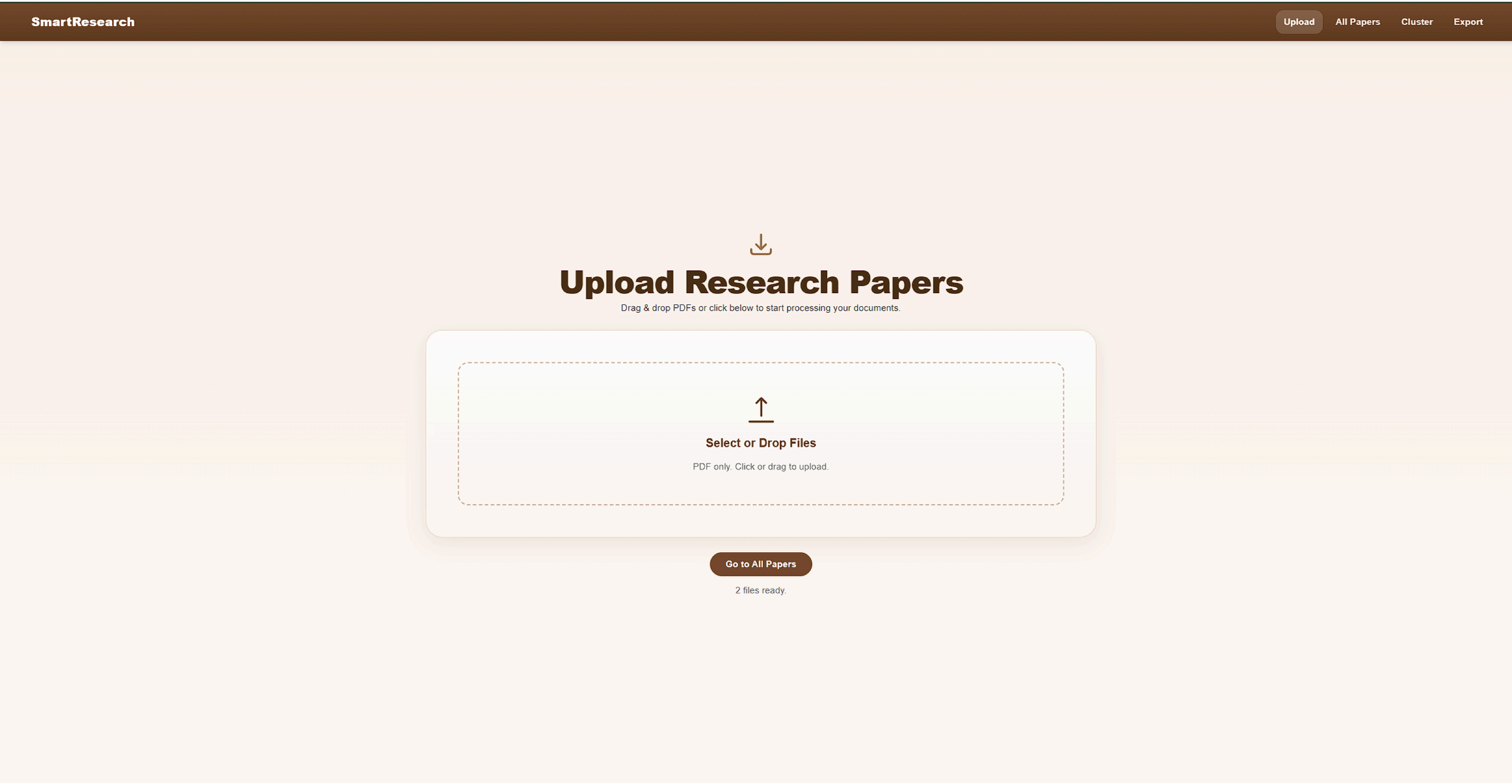
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Task:
You should write a program with your selected programming language (python, java, or C++) to simulate the queuing and service of bank customers. The simulation should be discrete event driven.

Your program should:
1. Read in the number of servers/tellers for the simulation,
2. Read in the file name from standard input and then open the file,
3. Read data record by record from the named file,
4. Perform the simulation according to the input records, and
5. Output the statistics of the services as specified below.

The input text file (**a1-sample.txt**) contains the following data:
1. A set of customers, one customer per line and each line consisting of a customer's arrival time, the corresponding service time, and the customer's priority (1 for low, 2 for normal, 3 for high).
2. The set of customers is sorted in the ascending order of their arrival times.
3. This set is terminated by a dummy record with the arrival time and the service time all equal to 0.

The simulation is to be a bank branch with **multiple tellers** (i.e., servers) and a **single queue**. The simulation should be run as follows:
1. Setup the total number of tellers;
2. Read the text file to start the simulation;
3. Allocate the new arrival customer to the next idle tellers (if applicable);
4. If all tellers are busy, add the new arrival customer to the queue;
5. The queue shall be updated based on two criteria of the customers, i.e., their arrival times and their priorities. For the customers with the same priority level, the customers who arrive earlier shall be put in front of the customers who arrive late. However, the customers with a higher priority shall be put in front of the customers with a lower priority regardless of their arrival times. For example, if we have three customers, i.e., c1 (2.8, 4.8, 2), c2 (1.5, 3.1, 2), and c3 (3, 5.6, 3), then their order in the queue should be 'c3, c2, c1', which means the service order for these three customers should be c3, c2, c1.
6. If a busy teller finishes a service request, then the teller will serve the first customer in the queue (the customer will be removed from the queue). If the queue is empty, then the busy teller will become idle;
7. The simulation should be run until the queue is empty and the last customer has left the system.

The simulation will run multiple times to gather statistics on the efficiency of the service with a different number of tellers on duty. You are required to run the simulation four times with the number of servers

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Graph neural networks: A review of methods and applications

Jie Zhou [a,1], Ganqu Cui [a,1], Shengding Hu [a], Zhengyan Zhang [a], Cheng Yang [b], Zhiyuan Liu [a,*], Lifeng Wang [c], Changcheng Li [c], Maosong Sun [a]

[a] *Department of Computer Science and Technology, Tsinghua University, Beijing, China*
[b] *School of Computer Science, Beijing University of Posts and Telecommunications, China*
[c] *Tencent Incorporation, Shenzhen, China*

ARTICLE INFO

Keywords:
Deep learning
Graph neural network

ABSTRACT

Lots of learning tasks require dealing with graph data which contains rich relation information among elements. Modeling physics systems, learning molecular fingerprints, predicting protein interface, and classifying diseases demand a model to learn from graph inputs. In other domains such as learning from non-structural data like texts and images, reasoning on extracted structures (like the dependency trees of sentences and the scene graphs of images) is an important research topic which also needs graph reasoning models. Graph neural networks (GNNs) are neural models that capture the dependence of graphs via message passing between the nodes of graphs. In recent years, variants of GNNs such as graph convolutional network (GCN), graph attention network (GAT), graph recurrent network (GRN) have demonstrated ground-breaking performances on many deep learning tasks. In this survey, we propose a general design pipeline for GNN models and discuss the variants of each component, systematically categorize the applications, and propose four open problems for future research.

1. Introduction

Graphs are a kind of data structure which models a set of objects (nodes) and their relationships (edges). Recently, researches on analyzing graphs with machine learning have been receiving more and more attention because of the great expressive power of graphs, i.e. graphs can be used as denotation of a large number of systems across various areas including social science (social networks (Wu et al., 2020), natural science (physical systems (Sanchez et al., 2018; Battaglia et al., 2016) and protein-protein interaction networks (Fout et al., 2017)), knowledge graphs (Hamaguchi et al., 2017) and many other research

neural networks for graphs. In the nineties, Recursive Neural Networks are first utilized on directed acyclic graphs (Sperduti and Starita, 1997; Frasconi et al., 1998). Afterwards, Recurrent Neural Networks and Feedforward Neural Networks are introduced into this literature respectively in (Scarselli et al., 2009) and (Micheli, 2009) to tackle cycles. Although being successful, the universal idea behind these methods is building state transition systems on graphs and iterate until convergence, which constrained the extendability and representation ability. Recent advancement of deep neural networks, especially convolutional neural networks (CNNs) (LeCun et al., 1998) result in the rediscovery of GNNs. CNNs have the ability to extract multi-scale localized spatial

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Graph Signal Generative Diffusion Models

Yigit Berkay Uslu, Samar Hadou, Sergio Rozada, S. S. Bidokhti, Alejandro Ribeiro

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We introduce U-shaped encoder-decoder graph neural networks (U-GNNs) for stochastic graph signal generation using denoising diffusion processes. The architecture learns node features at different resolutions with skip connections between the encoder and decoder paths, analogous to the convolutional U-Net for image generation. The U-GNN is prominent for a pooling operation that leverages zero-padding and avoids arbitrary graph coarsening, with graph convolutions layered on top to capture local dependencies. This technique permits learning feature embeddings for sampled nodes at deeper levels of the architecture that remain convolutional with respect to the original graph. Applied to...

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Graph Mixing Additive Networks

Maya Bechler-Speicher, Andrea Zerio, Maor Huri, M. V. Vestergaard, Ran Gilad-Bachrach, Tine Jess, Samir Bhatt, Aleksejs Sazonovs

• 2025 • 0 citations

We introduce GMAN, a flexible, interpretable, and expressive framework that extends Graph Neural Additive Networks (GNANs) to learn from sets of sparse time-series data. GMAN represents each time-dependent trajectory as a directed graph and applies an enriched, more expressive GNAN to each graph. It allows users to control the interpretability-expressivity trade-off by grouping features and graphs to encode priors, and it provides feature, node, and graph-level interpretability. On real-world datasets, including mortality prediction from blood tests and fake-news detection, GMAN outperforms strong non-interpretable black-box baselines while delivering actionable, domain-aligned explanations.

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Exploiting the Structure of Two Graphs With Graph Neural Networks

Victor M. Tenorio, Antonio G. Marques

IEEE Transactions on Signal and Information Processing over Networks • 2024 • 1 citations

As the volume and complexity of modern datasets continue to increase, there is an urgent need to develop deep-learning architectures that can process such data efficiently. Graph neural networks (GNNs) have emerged as a promising solution for unstructured data, often outperforming traditional deep-learning models. However, most existing GNNs are designed for a single graph, which limits their applicability in real-world scenarios where multiple graphs may be involved. To address this limitation, we propose a graph-based architecture for tasks in which two sets of signals exist, each defined on a different graph. We first study the supervised and semi-supervised cases, where the input is a signal on one...

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Graph-Based Explainable AI: A Comprehensive Survey

M. Bugueño, Russa Biswas, Gerard de Melo, Aalborg University, Denmark Gerard, DE Melo, RCExplainer SubGraphX

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Faster Convergence in Mini-batch Graph Neural Networks Training with Pseudo Full Neighborhood Compensation

Qiqi Zhou, Yanyan Shen, Lei Chen

Proceedings of the VLDB Endowment • 2025 • 0 citations

Graph Neural Networks (GNNs) have achieved remarkable success in various graph-related tasks. However, training GNNs on large-scale graphs is hindered by the neighbor explosion problem, rendering full-batch training computationally infeasible. Mini-batch training with neighbor sampling is a widely adopted solution, but it introduces gradient estimation errors that slow convergence and reduce model accuracy. In this work, we identify two primary sources of these errors: (1) missing gradient contributions from unsampled target nodes, and (2) inaccuracies in messages computed from sampled nodes. While existing methods largely focus on mitigating the second source, they often overlook the first, resultin...

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